

Deriving vowel reduction from a law governing human motion

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Abstract

Phonological patterns often parallel phonetic ones; for example, phonological vowel reduction, the categorical neutralization of vowel contrasts in unstressed syllables, parallels phonetic vowel reduction, the gradient, incomplete neutralization of contrasts among vowels of short duration (Barnes, 2006). Theories of the phonetics-phonology interface (Hayes 1999, Flemming 2001, Eischens 2022, i.a.) explain these parallels by deriving phonological patterns from phonetic boundary conditions, inviolable physical or cognitive constraints that create principled gaps in linguistic substance (Lindblom 1972). Which boundary condition do phonetic and ultimately phonological vowel reduction derive from? And by what mechanism does this boundary condition shape the phonology or its output?

I propose that phonetic vowel reduction is a consequence of Fitts' law (Fitts 1954), a constraint on almost all human motions (Schmidt et al. 1979, MacKenzie 2018) including articulatory gestures (Kuberski & Gafos 2021) which states that decreases in duration force nonlinear decreases in a movement's magnitude and/or accuracy. Vowels with short durations (e.g., in unstressed syllables) are thus articulated with smaller, less accurate gestures, leading to vowel reduction.

In a case study of Coratino (Italo-Romance, southern Italy; data from Bucci et al. 2019), I show that the filtering of phonological outputs by the phonetics (Eischens 2022) based on Fitts' law captures a pattern of categorical vowel centralization conditioned by both duration and segmental context. Fitts' law explains this pattern, while phonological accounts which only indirectly refer to phonetic substance can describe but not explain it. This analysis functions as a proof of concept for the use of explicit models of the phonetics-phonology interface to connect well-defined boundary conditions to phonological patterns.

1 Introduction

Vowel reduction is commonly described as a process in which stress conditions systematic differences in vowel quality, often leading to the neutralization of vowel contrasts in unstressed syllables. However, contrary to this unitary description, vowel reduction varies in apparent categoricity vs. gradience of the change and conditioning by abstract vs. concrete factors (e.g., stress vs. duration). In other words, assuming that the phonetics is gradient and concrete and the phonology is categorical and abstract, vowel reduction varies in whether it appears to be phonetic or phonological (Barnes 2006), as well as which vowel qualities change and how (Crosswhite 2000, 2004). For example, raising, centralization, combinations of these two, and the elimination of vowel qualities other than [i, a, u] are all attested.

There are parallels between gradient vowel reduction conditioned by concrete factors such as duration and categorical vowel reduction conditioned by abstract factors such as stress. A common desideratum of phonological theories is an explanation of how phonological patterns arise from phonetic processes, particularly boundary conditions, inviolable physical or cognitive constraints that create principled gaps in linguistic substance (Lindblom 1972). From this perspective, vowel reduction presents an opportunity to evaluate theories which seek to explain the parallelism between gradient and categorical processes. Should this parallelism arise from phonological learning of phonetic tendencies (Hayes 1999, Barnes 2006), from interactions at the phonetics-phonology interface (Eis-chens 2022), or because there is no separation between phonetics and phonology (Flemming 2001)?

I make two proposals. First, vowel reduction may arise from Fitts' law (Fitts 1954), a constraint on almost all human motion (Schmidt et al. 1979, MacKenzie 2018) including speech articulation (Kuberski & Gafos 2021). It is modeled as a nonlinear equation, $MT = a + b \log_2(2D/W)$, that describes the relationship between a movement's time (MT , i.e., duration), displacement (D), and target width (W , the reciprocal of accuracy). As movement time decreases, people must choose to either decrease displacement and produce smaller movements, or increase target width and decrease the accuracy of their movements. Fitts' law thus acts as a boundary condition on speech articulation.

Second, the parallelism between phonetic vowel reduction arising from Fitts' law and phonological vowel reduction can be modeled as categorical phonological changes conditioned indirectly by Fitts' law via an interaction at the phonetics-

phonology interface. This interaction is described via the Phonetically-Informed Candidate Selection model, in which phonological outputs are in part selected based on phonetic criteria (Eischens 2022). I argue that Fitts’ law is the relevant phonetic criterion for vowel reduction, and theories which combine explicit models of phonetic boundary conditions and the phonetics-phonology interface have greater explanatory power than purely phonetic or purely phonological theories otherwise would.

I support these claims with an analysis of vowel reduction in Coratino (Italo-Romance, southern Italy), which displays both raising and centralization of unstressed vowels; the latter is blocked if an adjacent consonant shares its place of articulation with the vowel (Bucci et al. 2019). Based on an original analysis of Bucci et al.’s data set, I show that while raising is gradient and centralization is categorical, both are conditioned by the movement time of the vowel gesture, and that the three-way relationship between movement time, displacement, and target width that Fitts’ law describes provides an explanatory mechanism for the blocking of centralization.

The paper is organized as follows. §2 consists of an overview of the typology and possible analyses of vowel reduction focusing on gradience and categoricity of vowel quality changes and abstract vs. concrete conditioning factors, categoricity and gradience as diagnostics for phonological and phonetic phenomena, and an introduction to Fitts’ law. In §3, I sketch a proposal for how Fitts’ law may be used to analyze vowel reduction. In §4, I examine vowel reduction in Coratino and propose an analysis based on the filtering of phonological outputs based on their Fitts’ law-compliant realizations. §5 is the conclusion.

2 Background

2.1 The typology and analysis of vowel reduction

The vast majority of vowel reduction patterns involve the neutralization of height contrasts, with the neutralization of other contrasts such as backness or nasality being rare and typically secondary to height neutralization (Barnes 2006: pp. 20, 72). While vowel reduction may thus appear to be relatively uniform in its effects, it varies in two major ways. First, the change in vowel quality may be gradient or categorical. Second, the change may be conditioned by continuous physical factors such as duration or abstract categorical factors such as (the absence of) stress. One interpretation of this empirical situation

is that vowel reduction may occur in the phonetics, in which case it is gradient and conditioned by concrete factors, or in the phonology, in which case it is categorical and conditioned by abstract factors. I discuss the variation in vowel reduction patterns and how that variation may be interpreted in the following two sections.

2.1.1 Gradient vs. categorical changes and conditioning factors

Changes in vowel quality may be gradient, resulting in incomplete neutralization between adjacent vowel categories, or they may be categorical, in which case the contrast between two vowel categories is erased completely. Changes in quality may also be conditioned by continuous physical factors (almost always meaning duration; Barnes 2006: p. 29, Crosswhite 2004: p. 197), or by abstract, categorical factors, prototypically the presence or absence of stress. Brazilian Portuguese (Major 1985) provides an example of categorical, stress-conditioned vowel reduction and gradient, duration-conditioned vowel reduction coexisting in a single language.

This language has seven vowel qualities, /i, e, ε, a, ɔ, o, u/. These vowels undergo two types of raising in unstressed syllables. The first type is the categorical raising of the mid-low vowels to mid-high in all unstressed syllables regardless of duration, resulting in a five-vowel inventory, [i, e, a, o, u]. The second type is the raising of non-high vowels in unstressed syllables as a function of duration. [e] and [o] raise to [i] and [u], resulting in a three-vowel inventory, [i, a, u]. Unlike the raising of the mid-low vowels, this raising is optional, gradient, and sensitive to the duration of the vowel. Processes that increase vowel duration (e.g., prepausal lengthening) decrease the likelihood that the vowels will raise, and processes that decrease vowel duration (e.g., posttonic shortening) increase the likelihood of raising.

/a/ may also raise slightly to [ɐ]. This raising appears to be correlated with both duration and posttonic position, though more strongly with the latter (Oh 2021); it is unclear whether it should be interpreted as gradient or categorical. Some examples of both categorical and gradient raising of mid vowels in Brazilian Portuguese are given in Table 1. In all tables with examples of vowel reduction, the relevant vowels are underlined.

Underlying	Stressed	Unstressed		
		Long duration	Short duration	
/e/		me'nino	mi'ninu	'boy'
/o/		po'lido	pu'lidu	'polite'
/ɛ/	'fɛsta	fɛs'tjivo	fɛs'tjivu	'festive'
/ɔ/	'kɔmi	kɔ'memus		'we eat'

Table 1: Obligatory, categorical raising of unstressed mid-low vowels and optional, gradient raising of mid-high vowels conditioned by duration in Brazilian Portuguese (Major 1985: pp. 266–7).

2.1.2 Phonetic vs. phonological analyses of vowel reduction

While the patterns of vowel reduction presented so far have been deliberately characterized in a relatively theory-neutral manner, it is clear that some vowel reduction patterns result in categorical changes in vowel quality and are conditioned by abstract properties such as whether a syllable is stressed or unstressed, while others result in gradient changes and are conditioned by physically continuous factors such as the surface duration of the vowel. Despite these differences, there is also a degree of isomorphism between these two types of vowel reduction. Vowel reduction tends to result in broadly similar changes in vowel quality: raising and/or centralization. It is conditioned by stress either directly or indirectly via the effect that the presence or absence of stress (in combination with other factors) has on a vowel's duration. And it often results in the neutralization of contrasts between vowel qualities, though that neutralization may be incomplete in the case of gradient vowel quality changes.

How should we decide whether to model a particular pattern of vowel reduction (or any process involving changes in speech sounds) in the phonology or the phonetics? One common way to make this decision is to identify categorical mental representations with phonology and continuous mental representations with phonetics, and then assume that the physical categoricity or gradience of a process directly reflects its mental representation as categorical or continuous, i.e., to treat physical categoricity or gradience as directly diagnostic of whether a process is phonetic or phonological (Bennett, Henderson & Harvey 2023: 407–8 and references therein). If we take this criterion at face value, then determining whether a phonetic or phonological analysis is appropriate for a given phenomenon should be as simple as checking the phonetic data for evidence of categoricity or gradience.

However, it is not safe to assume that speaker behavior straightforwardly

reflects mental representations. A process that is mentally represented as categorical but implemented noisily may appear physically gradient, while a process that is mentally represented as continuous but whose implementation contains a nonlinear component may appear physically categorical (Gafos & Benus 2006). In other words, even if we choose to define the phonology and phonetics in terms of categorical vs. continuous mental representations, respectively, the physical categoricity or gradience of a phenomenon is not necessarily a reliable diagnostic of how it is represented mentally.

Another way in which the decision to model a phenomenon as phonological or phonetic might be made is with reference to speaker control. I use the term “speaker control” to mean aspects of speaker behavior that cannot be reduced to an automatic process or random chance. In other words, speaker control describes anything which could in principle have been some other way but is not and which thus reflects a decision on the part of the speaker, whether that decision was made online or offline (e.g., during acquisition), consciously or unconsciously, etc. Many diagnostics which purportedly test for phonetic vs. phonological behavior, such as (in)sensitivity to speech rate (which can be understood in the context of vowel reduction as being about the surface duration of individual vowels), in fact diagnose speaker control (Bennett, Henderson & Harvey 2023: p. 442). Such diagnostics rely on the assumption that speaker control is exclusively a property of the phonology, while the phonetics is entirely automatic and not controlled.

Evidence for speaker control over fine phonetic detail suggests that a theory in which the phonetics is mechanistic and totally free from speaker control, or even one in which it is automatic but flexible and context-sensitive, is not tenable (Kingston & Diehl 1994). For example, patterns of nasal coarticulation may be language-specific (Cohn 1993, Solé 1995, Styler 2017) or even speaker-specific, and in at least some cases the differences between these patterns are not reducible to differences in phonological representations or the presence or absence of certain contrasts (Zellou 2022). Such phenomena are difficult to understand without some notion of speaker control over fine phonetic detail. If we accept that the phonetics is (at least partially) controlled, it is unsafe to assume that speaker control over a process entails that that process is phonological.

An example of a language in which a categorical change could be analyzed as phonological or phonetic is Standard Italian, in which /ɛ/ and /ɔ/ are realized as [ɛ] and [ɔ] in primary stressed syllables and as [e] and [o] in other syllables without any evidence of a gradient transition between these qualities (Baroni

1996 as cited in Crosswhite 2004: p. 202). Meanwhile, /a/ is gradiently raised to [e] in unstressed syllables as vowel duration decreases (Farnetani & Vayra 1991). The categoricity of the raising of /ε/ and /ɔ/ and the obvious presence of speaker control could indicate that the raising of [e, o] should be modeled as a phonological process, while the gradience and apparently automatic nature of /a/-raising may indicate that it should be modeled as phonetic.

However, it is also possible to make a phonetic generalization about the distribution of the categorical raising of /ε, ɔ/ to [e, o]: unstressed and secondary stressed syllables are similar to each other in duration but shorter than primary stressed syllables (Farnetani & Kori 1990). As Barnes (2006: p. 31) notes, a possible interpretation is to say that duration and not stress per se conditions the neutralization of the mid-high and mid-low vowels. In other words, there is categorical reduction of the mid vowels alongside gradient reduction of /a/, the former of which may be sensitive to surface duration and the latter of which is certainly a function of surface duration.

An explicit, predictive model of interactions between the duration of an articulatory gesture and its magnitude would be useful in understanding these phenomena. Ideally, this model would make testable predictions about the behavior of a process which differed depending on whether its physical categoricity or gradience and conditioning by abstract or concrete factors were attributed to the nature of its mental representation or to details of its physical implementation.

It would also be desirable for the articulatory component of such a model to apply both to articulation and to general human motion, so that insights about movements of other parts of the human body might be extended to speech production. Fortunately, such a model of human motion not only exists, but is well-established in the literature: Fitts' law.

2.2 Fitts' law

Fitts' law (Fitts 1954) is a constraint on human motion which describes the proportional relationship between three quantities: a movement's duration, displacement (distance crossed), and target width (accuracy criterion). This relationship is stated as an equation. The common version of this equation, taken here from Schmidt et al. (1979), is

$$(1) \quad MT = a + b \log_2 \left(\frac{2D}{W} \right)$$

where MT = movement time (duration), D = displacement, W = target width, a sets the absolute minimum movement time, and b is a scaling term. a and b are constants determined by empirical testing. The relationships that this equation describes are as follows. First, for a given duration, displacement and target width are proportional to one another such that an increase in one results in an increase in the other, i.e., larger movements are less accurate for the same duration. Second, for a given displacement, duration and target width are inversely proportional to one another, i.e., increasing duration results in greater accuracy for the same displacement. Third, for a given target width, duration and displacement are proportional to one another, such that an increase in one results in an increase in the other, i.e., larger movements take longer for the same accuracy.

In other words, humans face a tradeoff between speed and accuracy when executing their movements, and the shorter a movement is, the worse the tradeoff becomes.¹ This relationship is illustrated in Figure 1.

As can be seen in panels (a) and (b), D and W differ in how nonlinear their relationship with MT is and whether the majority of the change in D or MT occurs before or after the area of greatest nonlinearity in each curve.

Panel (a) shows that as displacement increases, so does movement time. However, the relationship is nonlinear. The majority of the increase in MT occurs in the transition from very small displacement to intermediate displacement; as D increases, the rate at which MT increases slows. For larger targets (greater values of W), the relationship between D and MT approaches linearity. Consequently, if D is being calculated as a function of MT , a decrease from long to intermediate MT results in a greater loss of D than a decrease from intermediate to a short MT .

Meanwhile, the curves in panel (b) shows more pronounced nonlinearity. For narrow targets (small values of W), MT is very long; as W increases, MT drops rapidly before reaching a plateau, at which point increasing W produces diminishing returns in terms of decreasing MT . For larger movements (smaller values of D), this nonlinearity is less pronounced, though still present. In other words, there is a speed-accuracy tradeoff whose properties are affected by D . If W is being calculated as a function of MT , changes in MT above the area of

¹While Fitts' law has been characterized as describing a speed-accuracy tradeoff (e.g., Lammert et al. 2018), the collapsing of duration and displacement into a single measure of *speed* obscures the three-way relationship between MT , D , and W . I wish to emphasize that MT is not an immutable input to motor planning but a parameter which can equally well be adjusted if necessary to maintain a certain minimum W and D .

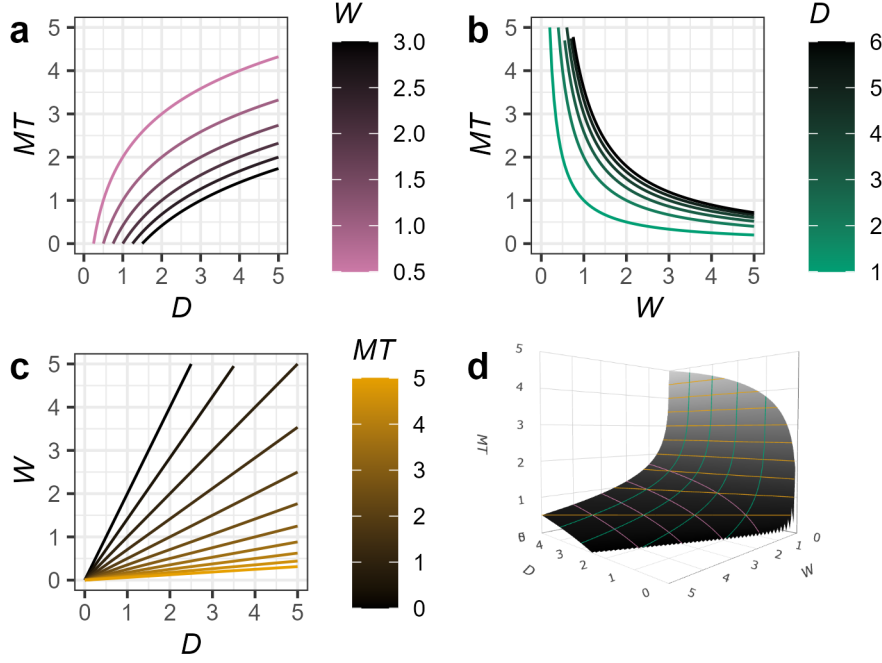


Figure 1: Solutions in dimensionless units for equation (1) given $a = 0$, $b = 1$. (a-c): solutions in 2-dimensional space. (d): 3-dimensional solution surface.

nonlinearity have only small effects on W . For example, for $D = 1$, small changes in MT lead to very large changes in W below a threshold of roughly $MT = 2$. This threshold is lower for smaller D , meaning that for small movements, the speed-accuracy tradeoff is not felt until very short MT , and then it is very nonlinear. For larger D , the threshold is higher and the overall curve closer to linear, meaning that larger movements encounter a closer-to-linear duration-accuracy tradeoff more quickly.

Panel (c) shows that the relationship between W and D is linear for any given MT , but the effect of MT on this relationship is nonlinear. With very long MT s, large movements may be almost arbitrarily accurate; the slope of the line is nearly horizontal. As MT decreases, the speed-accuracy tradeoff worsens at an accelerating rate; the slope begins diverging from horizontal more rapidly.

3 Vowel reduction as an articulatory consequence of Fitts' law

I propose to model phonetic vowel reduction as an effect of Fitts' law, and to characterize different patterns of phonetic vowel reduction as vowel articulations subject to reductions in displacement, increases in target width, or a combination of the two. In this section I sketch a model of articulatory targets with a W parameter and briefly discuss some ways in which speakers might respond to decreasing MT .

3.1 The nature of targets

So far I have left the notion of “target” underspecified. I adopt the notion of convex target regions proposed by Guenther (1995) in which a target is represented as an area within the speaker's mental representation of the vocal tract. For each articulator, the target region associated with a particular sound category is the smallest region which meets two criteria. First, the target region is convex, i.e., it has no inward-pointing corners. This convexity results from defining the region as the overlap of two intervals on non-parallel dimensions, e.g., a certain range of jaw heights with a certain range of tongue body backnesses. Second, the target region encompasses all points where the articulator may be placed that are compatible with the production of the sound in question. (How the speaker decides which points meet this criterion is a separate question from how Fitts' law gives rise to vowel reduction in the phonetics, and whose answer is outside the scope of this work. For now, I will simply assume that the speaker makes such a judgment for every vowel they produce and leave the method an open question.) The width of this target region on a given articulatory dimension is its value of W on that dimension, e.g., the range of acceptable jaw heights for a particular target region is that target's W on the jaw height dimension.

We can consider the center point of this region to be the “canonical” target value, with the rest of the target region being productions judged “close enough” under current speaking conditions. As a one-dimensional example, suppose that jaw height is valued from 0 (as closed as physically possible) to 10 (as open as physically possible), and the target region for the vowel /a/ is centered on a

²Though I have used slashes, the model sketched here is compatible with any abstract symbolic mental representation of a sound category, including both underlying forms and the

value of 5. If the speaker wishes to articulate an /a/, they must open their jaw to a degree that falls within the target region. If the region expands (i.e., if W increases), then a larger range of jaw heights is judged acceptable for the production of /a/.

Though a target region may encompass many possible realizations, a single one must actually be produced. How is it chosen? The original information-theoretic equations on which Fitts modeled his own equations of human motion assume that information is transmitted in a channel that is noisy in proportion to the signal's magnitude (MacKenzie 2018: p. 355). From a neurological perspective, Harris & Wolpert (1998) argue that the human nervous system is such a channel: larger and/or faster movements are less accurate than slower and/or smaller ones because neural control signals are noisy in proportion to their magnitude; therefore, larger and thus noisier control signals are required for larger and faster movements. From an information-theoretic perspective, as MT decreases, so does the amount of information that can be transmitted with each motor command; the larger and more accurate a movement is, the more information is required to specify that *that* is the desired outcome, and not some other movement. In other words, as a movement's MT decreases, the uncertainty in its outcome increases. W thus represents the maximum uncertainty that can be tolerated for a given movement (Fitts 1954: p. 387).

Consequently, as a first cut I make the simplifying assumption every given phonetic realization of a phonological feature is a random sample from a normal distribution whose standard deviation (SD) equals W and whose expected value (mean) is the canonical target value. Equivalently, W of a target region in a given context equals the expected value (the mean) of the distribution of realizations of the targeted sound in that context, plus or minus its SD . For example, if the distribution of jaw height for realizations of /a/ in stressed syllables where MT is high has a SD of .5, then W of the jaw height target region for /a/ in that context is the expected jaw height of /a/ $\pm .5$.

This sketch incorporating Fitts' law immediately makes one successful prediction. As Turk & Shattuck-Hufnagel (2020) point out, the proportional relationship between displacement and duration for a given target width may explain why low vowels tend strongly to have greater durations than high vowels (Toivonen et al. 2015 and references therein). Assuming equal W , low vowels require a jaw movement that is larger and thus longer in duration to achieve outputs of phonological computations, i.e., with phonemes or allophones.

when starting from a closed position, i.e., D is greater. Because MT is derived from the base-2 logarithm of $2D/W$, if W is held constant, then as D increases, so will MT , though the rate of increase will taper off due to the logarithmic transformation (see Figure 1a). The corollary of this is that as MT decreases, so must D if W is held constant; shorter durations should thus result in articulatory undershoot if target width does not change.

3.2 Predicted patterns of phonetic vowel reduction

Referring to Figure 1c, for a given MT the Fitts' law equation has infinitely many solutions representing a ratio of W to D (a line), rather than one pair of numbers (a point). Speakers can thus respond to changes in MT by decreasing D to maintain the same W (downward and leftward movement in Figure 1a), increasing W to maintain the same D (downward and rightward movement in Figure 1b), or any combination of adjustments to D and W that satisfies the equation (downward movement on the MT axis and movement in both the D and MT dimensions in Figure 1d). Combined with the assumptions made in §3.1, this predicts three qualitatively different forms that phonetic vowel reduction might take, which are illustrated in Figure 2 alongside the expected form of categorical vowel reduction.

Increasing target width while holding displacement constant would lead to increased variance in the realization of a category without a change in the mean of the realizations' distribution (Figure 2a). In this case, the average height of the vowel is not expected to change, but the variance in its height is expected to increase while movement time decreases.

Decreasing D while holding W constant would lead to a change in the mean of the realizations' distribution without a change in variance (Figure 2b). For jaw opening, the variance in vowel height is expected to stay the same while the displacement from the previous (and possibly upcoming) jaw height is reduced. In other words, because displacement is reduced, the jaw height with which the vowel is produced will be closer to the jaw heights necessary for the surrounding segments. Because these segments will most often be consonants, which are generally produced with a relatively raised jaw, reduced displacement will most often result in vowel raising. If the jaw and tongue body were to start relatively lower, then the result of reduced D should be vowel *lowering*, as the maximum upward displacement of the jaw and tongue body are limited. A possible example is the lowering effect that uvular and pharyngeal consonants

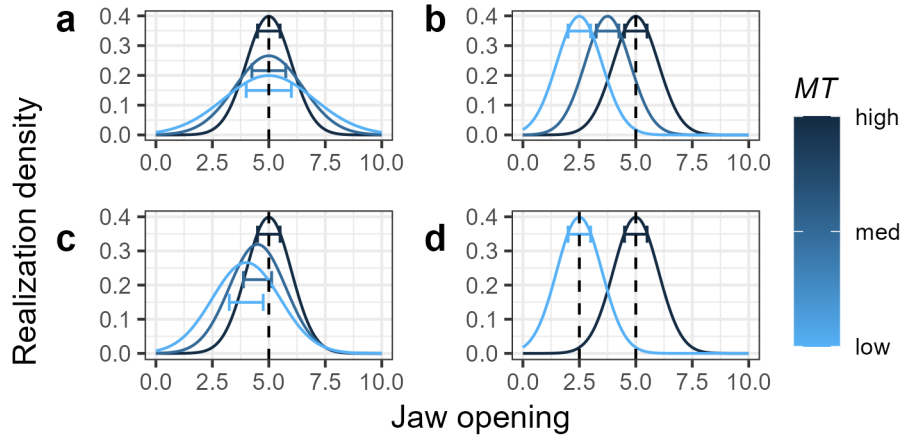


Figure 2: Hypothetical density plots of jaw height realizations for /a/ illustrating four responses to decreasing MT . Vertical lines represent canonical targets; horizontal lines represent W , which is equivalent to the expected value $\pm SD$. (a) Increase W (increased variance). (b) Decrease D (gradient raising). (c) Increase W and decrease D (gradient raising with increased variance). (d) Switch targets (categorical raising).

have on adjacent vowels in Kabardian (Choi (1991) as cited in Ladefoged & Maddieson 1996: p. 287).

Simultaneously increasing W and decreasing D would lead to both increased variance in realizations of a category and the movement of the mean of those realizations’ distributions, but to a lesser degree than with changes in only W or only D (Figure 2c). In terms of jaw opening, the distribution of realizations should appear to be “stretched” towards the jaw heights necessary for the surrounding segments as the mean of the distribution moves and its variance also increases.

There is some evidence that speakers do simultaneously increase target width and reduce displacement. Guenther (1995) describes a strategy for faster articulation in which speakers expand the target region while aiming for its edge rather than its center so that the displacement and movement time required to reach it is less. In effect, speakers ‘move the goalposts,’ stretching the target region such that it still approximates its canonical value, but is also within reach of movements that are limited in D and MT . Guenther points out that increasing speech rate leads to increases in the speed of consonantal gestures but

smaller increases or even decreases in the speed of vowel gestures, and argues that vowel targets may expand in both tongue body height and backness, but consonants have a relatively fixed degree of constriction (tongue body height) and can only expand in place (tongue backness).³ Thus, for vowels, speakers may substantially increase W in multiple dimensions to save on D and thus reduce the speed of their articulations, while for consonants the increase in W and savings on D are less.

The final response to decreasing movement time, illustrated in Figure 2d, is to switch to a different target entirely. In other words, phonetic vowel reduction is preempted by the substitution of a nearer or wider target for a more distant or narrower one. For jaw opening, this would result in the presence of two distinct distributions associated with different MT s, i.e., to result in allophonic variation in height that is conditioned by duration. It is possible that duration could directly condition an abstract, symbolic change; (Eischens 2022) presents evidence for speech rate conditioning changes in abstract phonological representations in San Martín Peras Mixtec.

If the original target and its substitute are associated with contrasting sound categories, for example $/e/$ and $/i/$, then switching targets will result in both a categorical change and a categorical neutralization between $/e/$ and $/i/$. On the other hand, if the two targets are *not* associated with contrasting sound categories, then the result is a categorical change without neutralization. For example, suppose that in a language which contrasts the vowels $/i, e, a, o, u/$, the jaw opening and tongue body height targets for $/a/$ are exchanged for targets otherwise associated with $/e/$ and $/o/$. The result would be a categorical change to $[ə]$, i.e., categorical but non-neutralizing raising.

³While this may be true for languages such as English in which consonants are realized with a relatively consistent degree of closure, it does not need to be universal. We can imagine languages that accept increased consonantal closure W in exchange for shorter D , i.e., greater variability in the degree of constriction and movements of smaller magnitude towards that closure, resulting in consonant lenition whose degree is proportional to MT and D . One could speculate that voiced stop spirantization in Spanish may have had its origins in such a process, and perhaps look for a connection between the presence of consonant lenition and tradeoffs between W variabilities in consonantal vs. vocalic gestures.

4 Case study: two patterns of vowel reduction in Coratino

An implicit assumption so far has been that speakers have a choice between the strategies described in §3.2 for each articulator or perhaps even each individual target. But are the predictions of Fitts’ law for vowel reduction actually borne out, and if so, to what extent do speakers systematically employ different strategies for each articulator, vowel target, or context?

In this section, I describe and analyze vowel reduction in Coratino, which is spoken in the town of Corato and the surrounding area in Apulia in Southern Italy. I show that two patterns of vowel reduction coexist in this language and analyze them as consequences of Fitts’ law and the relationship between duration, displacement, and target width that it imposes, either directly via gradient changes which occur in the phonetics or indirectly via phonetically conditioned phonological alternations.

Because this argument crucially depends on details about the phonetic realizations of unstressed vowels in Coratino, I discuss the phonetic facts in some detail in §4.3 and §4.4 before sketching an analysis in §4.6.

4.1 Background

Vowels in Coratino display systematic differences in quality across different prosodic and segmental contexts. Stressed syllables contain seven vowel qualities, /i, e, ɛ, a, ɔ, o, u/. Some examples are given in Table 2.

	Front	Central	Back
High	'viŋŋə 'vine'		'sukə 'juice'
Mid-high	'venə 'comes'		'sokə 'I am'
Mid-low	'vɛnə 'vein'		'sɔkə 'rope'
Low		'sakka 'sack'	

Table 2: Stressed vowels of Coratino with examples (D’Introno & Weston 2000: pp. 91, 93, Bucci et al. 2019: p. 316).

Vowels in unstressed syllables display qualitatively different patterns of vowel reduction which appear to be in complementary distribution across three types of position, which I refer to using terminology originally employed by Bucci et al. (2019). *Unstressed initial* vowels are found in absolute word-initial position. Non-high vowels raise slightly; it is unclear whether this results in a merger,

because the data set contains only a single observation of an initial underlyingly mid-high vowel. Some examples are given in Table 3.

Underlying	Stressed		Unstressed	
/i/	^ˈ itə	‘high’	itə ^ˈ tʃiddə	‘tall person’
/ɛ/	^ˈ ervə	‘grass’	ɛrvə ^ˈ tʃeddə	dim.
/a/	^ˈ andʒə	‘angel’	^ˈ aniddə	‘ring’
/ɔ/	^ˈ ɔrdənə	‘order’	ɔrdə ^ˈ na	‘ranger’
/u/	^ˈ uʃʃə	‘oil’	^ˈ uʃja	‘oiled’

Table 3: Mid-low vowels raise slightly in unstressed word-initial position; mid-high vowels, stressed or unstressed, are rare or unattested in word-initial position (adapted from Bucci et al. 2019: pp. 291, 316).

Unstressed protected vowels occur adjacent to a consonant with a shared place of articulation. For front vowels, this is a palatal consonant; for back vowels, this is either a labial or velar consonant. It does not appear to matter whether this consonant precedes or follows the vowel, e.g., [bul^ˈlinə] ‘stamp (dim.)’, [fiʃjettə] ‘girl (dim.)’, or whether the consonant and vowel are in the same syllable, e.g., [lu^ˈminə] ‘light (dim.)’. Non-high unstressed protected vowels are raised to a greater degree than in unstressed initial position; Bucci et al. (2019) report that /ɛ/ is raised to [e] adjacent to palatals, /ɔ/ is raised to [o] adjacent to labials, and both /ɔ/ and /o/ are raised to [u] adjacent to velars. No category of consonants systematically influences the realization of /a/; consequently, I do not include /a/ in the discussion of unstressed protected vowels. Some examples are given in Table 4.

Underlying	Stressed		Unstressed		Shared place
/i/	^ˈ vɪɲə	‘vine’	vɪɲə ^ˈ təddə	dim.	palatal
/e/	^ˈ cesə	‘church’	cesə ^ˈ rəddə	dim.	palatal
/ɛ/	^ˈ nɛʃʃə	‘fog’	nɛʃjʊsə	‘foggy’	palatal
/ɔ/	^ˈ pɔvərə	‘poor’	pɔvə ^ˈ riddə	dim.	labial
	^ˈ kɔndʒə	‘tint’	kundʒə ^ˈ torə	‘tanner’	velar
/o/	^ˈ mollə	‘elastic’	mɔl ^ˈ lettə	dim.	labial
/u/	^ˈ lume	‘light’	lu ^ˈ minə	dim.	labial

Table 4: Raising of unstressed protected mid-low vowels to mid-high, or raising of back vowels to [u] adjacent to velars (Bucci et al. 2019: p. 291).

Unstressed nonprotected vowels are unstressed vowels that are neither initial nor protected, i.e., they occur elsewhere. Nonprotected non-high vowels raise to a similar degree as protected vowels. They also undergo centralization and loss

of rounding: /i/ and /u/ merge to [i], while /ɛ, ɔ/ merge to [ə].⁴ (/e/ and /o/ are either not attested or not distinguished from /ɛ/ and /ɔ/ in this position in Bucci et al.’s data set). It has previously been claimed that other unstressed vowels have just two qualities, [ə] and [a] (D’Introno & Weston 2000). However, Bucci et al. (2019: p. 303) report that the F1 value of centralized underlyingly high vowels (i.e., the F1 value of [i]) is consistently 50–75 Hz lower than centralized underlyingly mid vowels, and this difference is statistically significant. Some examples of this raising and centralization are given in Table 5.

Underlying	Stressed		Unstressed	
/i/	ˈrɪkkə	‘rich’	rɪkˈkonə	dim.
/u/	abˈbruːfə	‘you burn’	abbɹɪˈfa	‘to burn’
/e/	ˈsellə	‘seat’	səlˈlinə	dim.
/o/	ˈrɒtə	‘wheel’	rəˈtɛddə	dim.
/ɛ/	ˈfɛmənə	‘woman’	fəmənˈnɛddə	dim.
/ɔ/	basˈtɒnə	‘stick’	bastənˈtʃiddə	dim.
/a/	ˈtappa	‘plug’	təpˈpa	‘to plug’

Table 5: Reduction of unstressed vowels to a vertical three-vowel system in Coratino (Bucci et al. 2019: p. 290)

The patterns shown so far are summarized in Figure 3. It can be seen that the non-high vowels raise in unstressed syllables; the degree of raising is greatest in protected and nonprotected vowels, and intermediate in initial vowels. Meanwhile, nonprotected vowels undergo centralization. I show that these patterns of vowel reduction can be understood as the combined effects of i) a process of gradient raising which is conditioned by duration (§4.3) and ii) a process of categorical centralization which is conditioned by both duration and segmental context (§4.4).

4.2 Data source and methods of analysis

The data used in this study were originally presented in Bucci et al. (2019). The discussion in this section is based partly on their analysis and partly on an analysis original to the present paper which makes use of their data set, which they generously shared upon request. The data set consists of 1864 ob-

⁴Bucci et al. (2019) transcribe both high central and mid central vowels as [ə] in their Tables 1 through 4, from which many examples in this section are drawn, but differentiate them in later transcriptions. Based on their description of the pattern of vowel reduction which yields centralized vowels and my own analyses of their data I differentiate [i] and [ə] in all transcriptions.

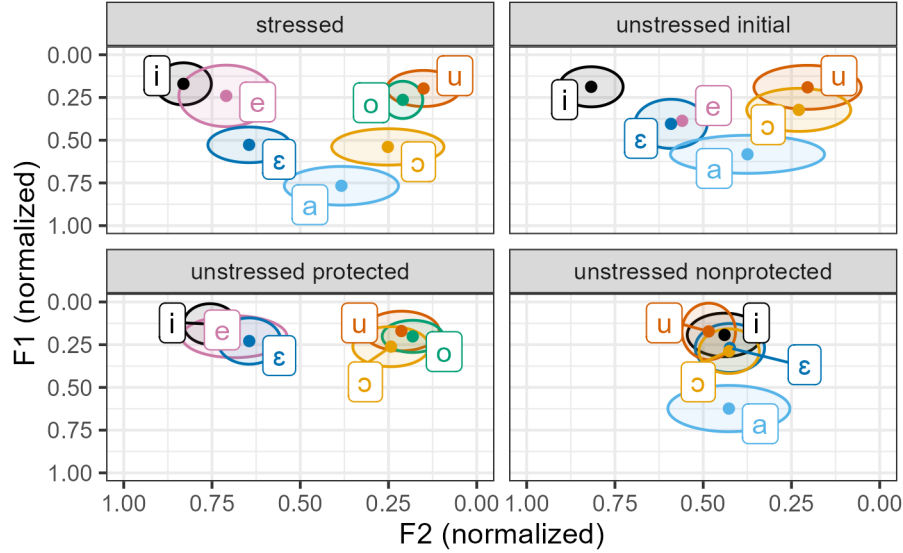


Figure 3: Mean values for realizations of each vowel in different contexts in normalized F1-F2 space. Points denote means; ellipses denote 1 standard deviation of the mean. Unstressed initial /e/ lacks an ellipse because there was only one observation.

servations of vowels extracted from words produced in isolation by 10 native speakers of Coratino (7 men, 3 women, ages 25 to 70). Each observation includes the speaker, the vowel’s underlying quality, its total surface duration in milliseconds, F1 and F2 measurements in Hertz taken from its midpoint, and the phonological context in which it appears (see §2). The underlying quality of vowels was determined by comparing related forms which displayed stress alternations, often but not always differentiated by the addition of a diminutive suffix as in the pair [ˈrotə], [rəˈtɛddə] ‘wheel (dim.)’. See Bucci et al. (2019: §2.2) for details on the methods of data collection and processing.

All data analysis original to this work was conducted in R (R Core Team 2023). In order to make different speakers’ productions comparable, F1, F2, and duration were normalized. Using the `rescale()` function on default settings (Wickham, Pedersen & Seidel 2023), for each speaker, the lowest recorded value of each phonetic measure was converted to 0 and the highest to 1, and all intermediate values rescaled to a value between 0 and 1 such that their relative distance from the lowest and highest values was maintained. For example, Speaker 1’s productions ranged between 34 and 262 ms in length; a duration of

148 ms (precisely halfway between 34 and 262 ms) would be rescaled to .500, while durations of 100, 150, and 200 ms would be rescaled to .289, .509, and .728, respectively. Consequently, when interpreting plots in this section, a value of 0 indicates the smallest measured value for each speaker, not 0 Hz or 0 ms, and two-place decimal values indicate percentiles.

Unless otherwise stated, all t-tests were two-tailed Welch’s (unequal variance) t-tests with a 95% confidence interval which were run on normalized measures pooled across speakers. In plots, mean F1 and F2 values for each vowel are labeled according to the underlying vowel quality rather than surface realization.

4.3 Gradient raising of unstressed vowels

Underlying / ϵ , a, ɔ / were raised in unstressed syllables. The F1 of underlyingly low and mid-low vowels in unstressed syllables covaried linearly with vowel duration in the bottom 60 percentiles of duration (Figure 4). In other words, the degree of raising corresponded linearly with duration when duration was below a threshold of roughly 175 ms (the mean duration of vowels in the 55th to 65th percentiles of duration, $SD = 25$ ms).

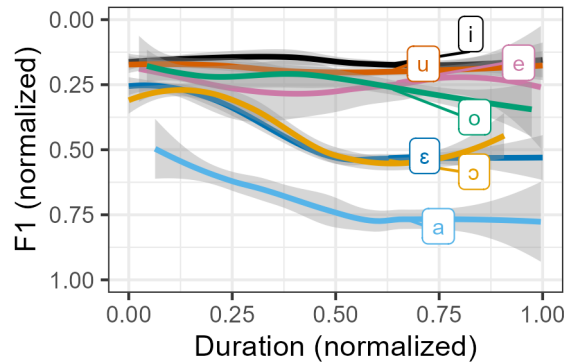


Figure 4: Smoothed lines for F1 of each vowel vs. duration, both normalized. Shaded areas represent a confidence interval of 1 standard error. Line fitting was performed using the locally estimated scatterplot smoothing (LOESS) method implemented via `geom_smooth()` function on default settings (Wickham 2016).

While / ϵ , ɔ / were raised to the point that their F1 approached values associated with stressed /e, o/, there is evidence that speakers did not merge these categories in their productions. Assuming that statistically significant differ-

ences in F1 correspond to a successfully realized difference between two height categories, if the height contrasts among non-low vowels had reduced from three categories (high, mid-high, and mid-low) to just two (high and mid) then we should find statistically significant differences in F1 between the groups that have remained separate, but no significant difference between the groups that have been merged. In other words, the trimodal non-low height distribution should reduce to a bimodal one. For example, if there were a categorical neutralization of /i, e/ $\rightarrow$ [i], we should find statistically significant F1 differences in comparisons of /i/ vs. / ϵ / and /e/ vs. / ϵ /, but not /i/ vs. /e/; only two distributions, one corresponding to both /i/ and /e/ and one corresponding to / ϵ /, should remain.

This was not the pattern observed in the data for any set of heights. As can be seen in Figure 5, there is no context in which two height distributions have the same mean or completely overlapping distributions.

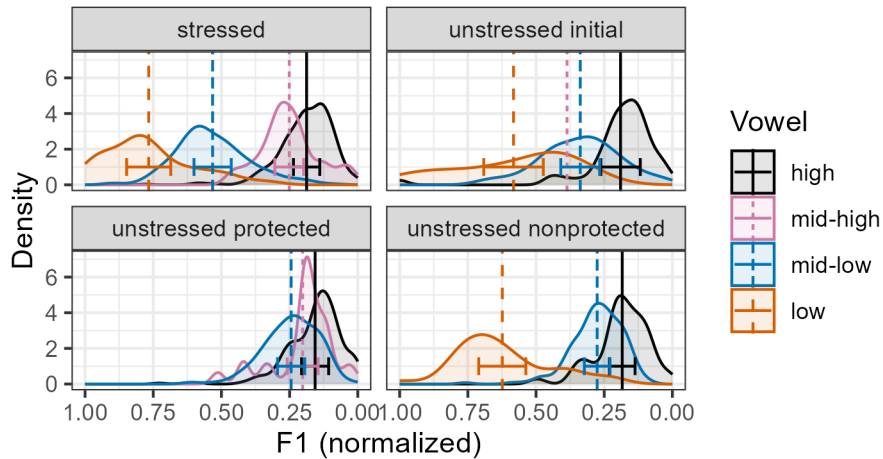


Figure 5: Density plot of normalized F1 for vowels of different heights in each context. Vertical lines denote means; horizontal bars denote 1 standard deviation of the mean.

Examining the distribution of normalized F1 in unstressed syllables across the bottom 40 percentiles of duration in Figure 6, it can be seen that the F1 distributions of underlyingly high, mid-high, and mid-low vowels partially overlap. Crucially, this overlap does not appear abruptly at a particular duration or in a particular unstressed context. Rather, it is present in all unstressed contexts and at all durations below the 40th percentile.

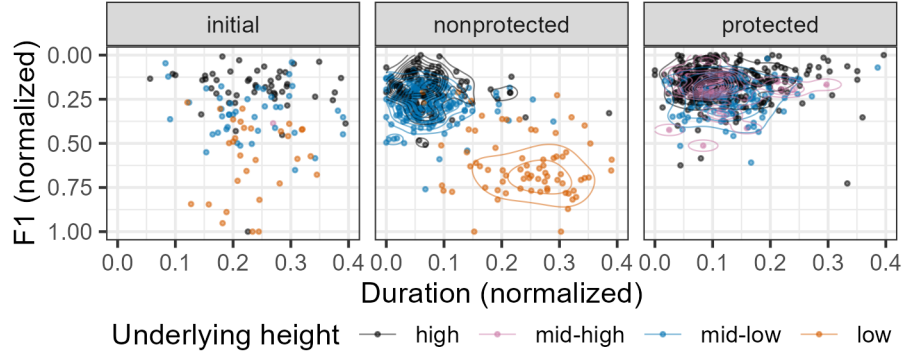


Figure 6: Density and scatter plots of normalized F1 in the bottom 40 percentiles of duration for unstressed vowels. A density function for vowels in initial position could not be calculated.

In other words, comparing F1's relationships with duration (Figure 4), with context (Figure 5), and with both simultaneously (Figure 6), changes in F1 (i.e., vowel raising) seem to be linked to changes in duration, not different contexts. In addition to this impressionistic description, t-tests were conducted for each pair of adjacent height distributions to determine if the difference between their means was significant. Means and standard deviations for each group are reported in Table 6.

Context	Height	<i>n</i>	F1	SD
stressed	high	444	.188	.097
stressed	mid-high	72	.251	.106
stressed	mid-low	304	.532	.137
stressed	low	113	.767	.162
initial	high	54	.190	.145
initial	mid-low	38	.338	.143
initial	low	30	.582	.218
protected	high	290	.156	.099
protected	mid-high	27	.202	.113
protected	mid-low	114	.244	.100
nonprotected	high	98	.184	.094
nonprotected	mid-low	189	.276	.094
nonprotected	low	79	.624	.173

Table 6: Counts, mean normalized F1, and standard deviations for vowels of different underlying heights in each context.

In stressed position, there were significant differences in mean normalized F1

between underlyingly high and mid-high vowels ($t(91.2) = 4.8$, $p < .001$, 95% CI: .090–.037) despite the fact that they overlap somewhat in Figure 3a. As expected, there were also significant normalized F1 differences between underlying mid-high and mid-low vowels ($t(133.2) = 19.07$, $p < .001$, 95% CI: .310–.252) and between underlyingly mid-low vowels and /a/ ($t(176.8) = 13.7$, $p < .001$, 95% CI: .201–.268).

Also as expected, there were significant differences in mean normalized F1 in unstressed initial position when comparing underlyingly high and mid-low vowels ($t(98.9) = 5.2$, $p < .001$, 95% CI: .204–.091) and underlyingly mid-low vowels and /a/ ($t(44.6) = 5.5$, $p < .001$, 95% CI: .154–.335). (Only one unstressed initial mid-high vowel was observed, precluding the calculation of a standard deviation and any statistical comparisons involving mid-high vowels in this position.)

Nonprotected vowels also showed statistically significant differences in normalized F1. High vowels and mid vowels differed significantly ($t(196.4) = 7.8$, $p < .001$), as did mid and low vowels ($t(97.9) = 16.9$, $p < .001$). These results agree with those obtained by Bucci et al. (2019: p. 303) using a χ^2 test; they find that for non-low nonprotected vowels, the effect of underlying vowel quality on F1 is significant ($\chi^2(5) = 75.2$, $p < .001$), hence the separate transcriptions of [i] and [ə].

Unstressed protected position is the only unstressed position in which high, mid-high, and mid-low vowels were recorded. Statistically significant differences in normalized F1 were found in comparisons of high vowels and mid-low vowels ($t(205.8) = 8.0$, $p < .001$, 95% CI: .156–.244), but not in comparisons of high vowels and mid-high vowels ($t(29.8) = 2.0$, $p > .05$) or mid-high vowels and mid-low vowels ($t(36.1) = 1.8$, $p > .05$). In other words, the distribution of non-low vowel heights was trimodal rather than bimodal, as can be seen in Figure 5c. While the space between the high, mid-high, and mid-low vowels has been compressed such that the mid-high vowels cannot be clearly differentiated from the high and mid-low vowels, the latter two remain differentiable from each other. There is thus no evidence of a categorical change in vowel height, i.e., two of the distributions being significantly different from the third but not from each other. I interpret this to mean that speakers retained separate high, mid-high, and mid-low targets in protected position and attempted to realize each separately.

No statistical evidence was found for categorical mergers in vowel height having taken place; speakers appeared to maintain all underlying height targets

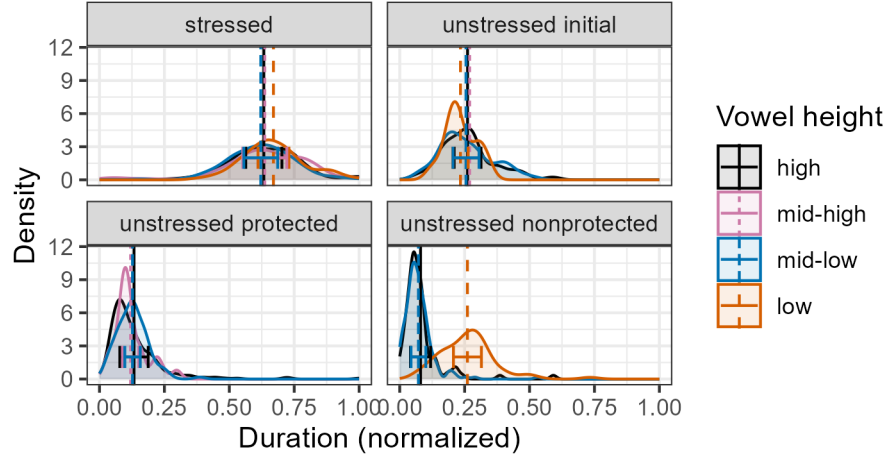


Figure 7: Distribution of duration measurements by context and vowel height. Vertical lines denote means; horizontal bars denote 1 standard deviation of the mean.

rather than replacing one target with another, even if raising resulted in the realizations of those targets partially overlapping. The degree of raising appears to be correlated with the duration of the vowel; consequently, I proceed with the assumption that F1 varied with duration rather than context. From this perspective, the fact that unstressed initial vowels were raised to a degree that is intermediate between no raising and that of unstressed vowels appears attributable to the fact that unstressed initial vowels were also intermediate in duration between stressed and other unstressed vowels.

In the following discussion, I infer the displacement of vowel gestures from a closed position based on their F1; greater F1 in a vowel is assumed to require a vowel gesture with greater displacement. The average degree of raising in each context is predictable from the average vowel duration in that context. Stressed vowels were the longest and showed no raising; non-initial unstressed vowels were the shortest and showed the greatest degree of raising; and initial unstressed vowels were intermediate in both duration and raising. Comparing the distributions of duration and F1 of each set of vowel heights in Figures 7 and 3, the same ordinal relationship is apparent: for both duration and F1, stressed > initial > protected $\approx$ nonprotected. This mirrors the trend shown in Figure 4a in which F1 decreases as duration does.

How should this be interpreted in terms of speaker responses to decreasing

vowel duration/movement time? Coratino speakers did not switch articulatory targets, as there is evidence of a gradient change in F1 rather than the nonlinearity associated with a categorical change. They also did not staunchly maintain displacement at the expense of increased target width: the means of each F1 distribution fell as duration/ MT decreased, with the exception of /a/, which I will return to. For underlyingly non-low vowels, it appears that Coratino speakers reduce displacement, allowing the distribution of their realizations to move closer to the values of adjacent segments, in this case, consonants produced with a relatively high jaw and tongue body; in other words, non-low vowels are raised (Figure 4). The lower the duration/ MT , the greater the loss of displacement. Based on the SD values in Table 6, the variance in F1 of mid-low vowels is smaller in protected position ($SD = .100$) and nonprotected position ($SD = .094$) than stressed position ($SD = .137$); this can also be seen in the error bars in Figure 5. In other words, the F1 variance of mid-low vowels decreases as they raise, suggesting that the displacement of these vowels decreases fast enough to not only counter the reduction in duration/ MT but also allow for a decrease in target width.

More generally, the variance in F1 of non-low vowels did not increase in unstressed positions, where duration was shorter, with the exception of word-initial position. Examining Table 6, vowels showed the greatest variation in F1 in initial position, despite the fact that initial vowels were intermediate in duration between stressed vowels and non-initial unstressed vowels (Figure 7). In terms of Fitts' law, it appears that speakers prefer to increase target width and decrease displacement simultaneously. At present, I do not have an explanation as to why initial position appears to differ from other unstressed positions in this way.

/a/ behaves differently from the non-low vowels. The mean normalized F1 of /a/ decreases from the 77th percentile in stressed syllables to roughly the 60th percentile in all unstressed syllables, i.e., there is a limit on how much /a/'s displacement can be reduced. /a/ also resists reduction of its duration; in nonprotected syllables, its mean duration remains at the 25th percentile while the mean of the other vowels' durations drops to roughly the 7th percentile. /a/ also consistently has a greater level of variance in its F1 distribution in unstressed syllables than vowels of other heights do in the same context, especially in nonprotected position ($SD_{\text{low}} = .173$, $SD_{\text{mid-low}} = .094$, $SD_{\text{high}} = .094$).

In sum, I interpret this to mean that i) speakers did not change targets at any point, because there was no evidence of categorical mergers; ii) for non-

initial non-low vowels, speakers respond to decreasing MT by decreasing D while maintaining W , because variance (standard deviations) did not increase as duration decreased; and iii) for /a/, speakers may both increase W and decrease D , but only to a certain maximum W and minimum D , effectively instantiating a minimum MT .

4.4 Categorical centralization of unstressed nonprotected vowels

Changes in F2 also correlated with duration, but in a context-sensitive fashion. The F2 of all vowels remained stable, or for /i/, declined gradually and slightly, in the upper 60 percentiles of duration. Below the 40th percentile, or roughly 125 ms (the mean duration of vowels in the 35th to 45th percentiles of duration, $SD = 18$ ms), the behavior of F2 differed by context. Means and standard deviations for each group are reported in Table 7.

Context	Backness	n	F2	SD
stressed	front	370	.732	.151
stressed	central	114	.383	.113
stressed	back	449	.185	.115
initial	front	17	.683	.171
initial	central	30	.375	.111
initial	back	86	.216	.127
protected	front	166	.712	.135
protected	back	265	.218	.116
nonprotected	front	178	.429	.138
nonprotected	central	79	.427	.135
nonprotected	back	109	.449	.147

Table 7: Counts, mean normalized F2, and standard deviations for vowels of different underlying backnesses in each context.

In an analysis including protected vowels and excluding nonprotected vowels, F2 of all vowels except /e/ remained stable across durations (Figure 8a). In an analysis including nonprotected vowels and excluding protected vowels, F2 of all non-central vowels (except /o/, for which there were not enough observations to calculate a trend) diverged from the 50th percentile as duration increased (Figure 8b). In other words, unstressed nonprotected vowels were centralized in a duration- and context-dependent fashion.

Contrary to what Figure 8b might suggest, this centralization appears to be categorical. Examining the distribution of F2 across the bottom 40 percentiles

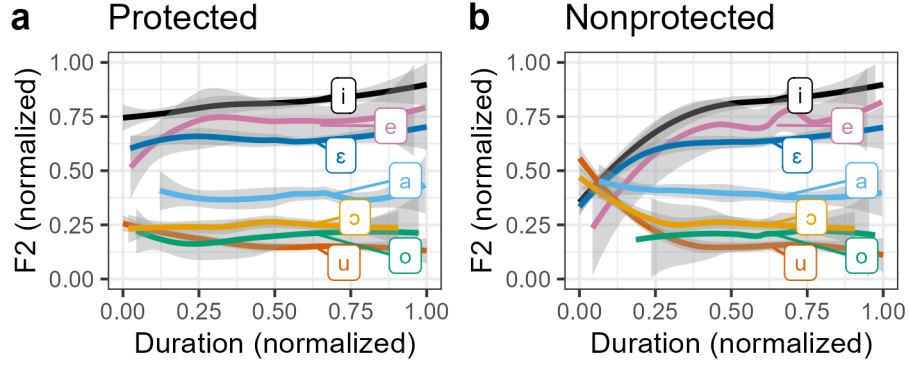


Figure 8: Smoothed lines for F2 of each vowel vs. duration, both normalized, in a subset of contexts. (a) Stressed, initial, and protected vowels. (c) Stressed, initial, and nonprotected vowels. Shaded areas represent a confidence interval of 1 standard error. Line fitting was performed as in Figure 4.

of duration for underlyingly non-central vowels in unstressed syllables, it is apparent that the distribution is bimodal in protected contexts and unimodal in nonprotected ones (Figure 9). This clear difference in modality across contexts is strikingly unlike the continuous change in F1 associated with decreasing duration, and suggests that rather than stretching a target region as duration decreases, speakers are switching from non-central targets to a central target.

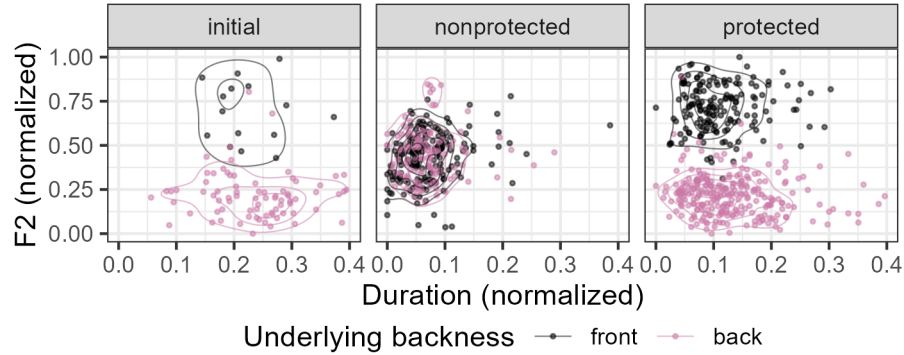


Figure 9: Density and scatter plots of normalized F2 in the bottom 40 percentiles of duration for underlyingly non-central unstressed vowels.

The F2 distribution in initial contexts is also suggestive of such a transition between a bimodal and unimodal distribution. Rather than a gradual change

in F2 which is proportional to the change in duration, non-peripheral F2 values suddenly appeared when duration fell below the 20th percentile. Further, front F2 values persisted even into the 10th percentile of duration.

No significant difference in normalized F2 was found between nonprotected underlyingly front and back vowels ($t(217.9) = 1.1$, $p > .05$, 95% CI: $-.014-.054$), but significant normalized F2 difference were found between vowels in stressed and nonprotected positions (front: $t(377.5) = 23.3$, $p < .001$, 95% CI: $.277-.328$; back: $t(142.0) = 17.5$, $p < .001$, 95% CI: $.234-.294$), as well as in protected and nonprotected positions (front: $t(341.2) = 19.1$, $p < .001$, 95% CI: $.253-.311$; back: $t(165.7) = 14.7$, $p < .001$, 95% CI: $.200-.262$).

I interpret this to mean that around the 20th percentile of duration, speakers began encountering difficulty achieving acceptable displacement and target width for backness targets. Instead of compromising on either, they chose to switch to a target that was achievable with smaller displacement, leading to a sudden change in the distribution of F2 that was conditioned by both duration and segmental context.

How can we distinguish this from simple reduction in displacement? If speakers respond to decreasing duration/ MT by sacrificing displacement, the change in D should be proportional to the change in MT (2b). In articulatory terms, we should expect gradient changes in vowel quality which correlate with duration. If speakers instead switch from one target to another, the change in D should be abrupt and not proportional to MT , though it might correlate with a particular durational/ MT threshold (Figure 2d). In articulatory terms, we should expect an abrupt change from one vowel quality to another.

The second pattern is what we see in the distributions of F2 in unstressed syllables. Rather than a gradual change in F2 values which correlates with changes in duration, the majority of nonprotected vowels displayed F2 values similar to those of /a/. Indeed, the distribution of F2 values for vowels of all underlying backnesses in nonprotected position was extremely similar (Figure 10d), suggesting that speakers attempted to hit only one backness target (the central one) in this context.

In this interpretation, what the protected and nonprotected contexts have in common is that only a small displacement is required to achieve the tongue backness target leading to the observed F2 distributions. The difference between these contexts is whether the canonical tongue backness target of the vowel is near that of an adjacent consonant. If the two target values are close enough together that the target value of the first movement is inside the target region

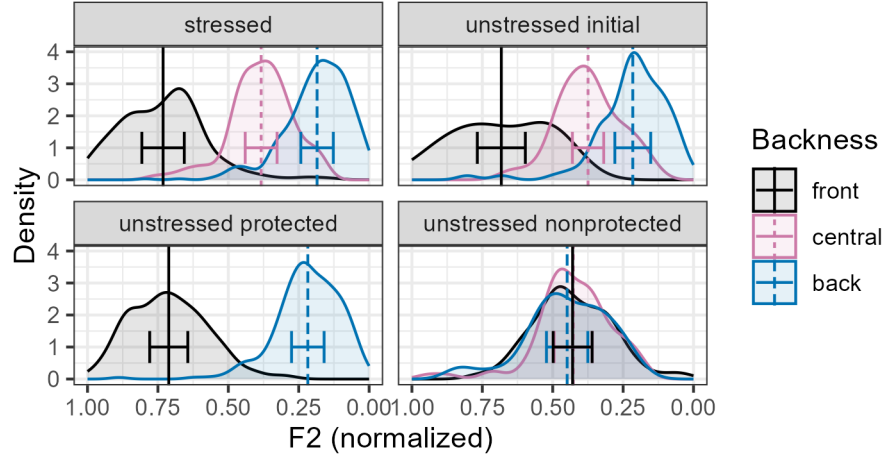


Figure 10: Density plot of normalized F2 for vowels of different heights in each context.

of the next, then one movement is sufficient to achieve the targets of both segments. Consequently, the consonant and vowel can be modeled as sharing a single tongue body backness gesture whose movement time is the sum of both segments' durations.

This interpretation is compatible with the way overlapping target regions must be handled in the Fitts' law equation $MT = a + b \log_2(2D/W)$ if we assume that (because movements with no or negative duration are not possible) MT must be greater than 0 for a solution to the equation to meaningfully describe an actual movement. Solutions in which displacement is less than half of target width (adjusted for the constants a and b) result in MT s of 0 or less; this can be shown to rearranging $a + b \log_2(2D/W) > 0$ (values of a , b , D , and W that result in MT s greater than 0) to produce the equation $D > W/2^{1+a/b}$ (values of D that result in MT s greater than 0 for a given set of a , b , and W). Thus, the Fitts' law equation cannot meaningfully describe a movement whose target region contains its starting point, and an adjacent segments with a shared target on some articulatory dimension *must* be modeled as sharing a single movement on that dimension; it is not possible to model this situation as a movement to the target followed by a second “movement” within the target region.

Returning to the case of protected vowels, I analyze them as having a tongue body backness gesture whose MT is longer than the vowel's duration by virtue

of sharing a target with an adjacent consonant. In this view, centralization really is exclusively conditioned by an *MT* in roughly the 20th percentile of duration or below (the percentile at which centralization begins to occur; see Figure 8), and nonprotected vowels are the only ones which have such low *MT*s.

Another indication that the *MT*s of protected vowels are longer than the duration of the vowels themselves is that protected and initial vowels pattern together in terms of their F2 distributions (see Figures 9 and 10), presumably because their tongue body backness gestures' *MT*s are intermediate between those of nonprotected and stressed vowels.

4.5 Interim summary

Based on an original analysis of Bucci et al.'s (2019) data set, I have argued that Coratino displays two types of vowel reduction, raising and centralization. Both are conditioned by decreases in *MT*, but beyond that show different characteristics.

For raising, *MT* is simply the vowel's duration, and speakers maintain four separate jaw/tongue body height targets. They accept some decreases in *D* and in the case of /a/ increases in *W*, resulting in gradient raising which covaries with duration.

For centralization, *MT* is a function of the vowel's duration and the segmental context. If the vowel shares a tongue backness target with an adjacent consonant, there is one tongue backness movement for both, and its *MT* is the sum of both segments' durations. Otherwise, *MT* is simply the vowel's duration. Speakers maintain separate front, central, and back target regions, and do not allow for large changes in *D* or *W*. Consequently, when *MT* becomes too low to hit front or back targets with acceptably large *D* and small *W*, speakers abruptly switch from front or back to a central target region.

In combination, these two patterns result in the following reductions. [i, e, ε, a, ɔ, o, u] with long *MT* gradually transitions to [i, e, ε, ɐ, ɔ, o, u] and then [i, e, ɐ, o, u] as *MT* decreases, i.e., height contrasts among non-low vowels are gradually and incompletely neutralized as speakers maintain all jaw/tongue body height targets but do not always reach the canonical value. When *MT* drops below a threshold corresponding to roughly the 20th percentile of duration, there is an abrupt transition to [i, ə, ɐ] in which backness contrasts are categorically neutralized as speakers switch tongue backness targets from non-central to central (Figures 8, 9, and 10). Effective *MT* can be increased

beyond the duration of the vowel if the vowel’s tongue backness target region overlaps that of an adjacent consonant, meaning that *MT*s low enough to trigger centralization are only seen in vowels that do not share a place of articulation with an adjacent consonant.

A final question remains. Why are initial vowels intermediate in duration between stressed and other unstressed vowels? Word-initial segments are often longer than their non-initial counterparts (White & Turk 2010, White 2014). Initial lengthening may also partially obviate duration-based vowel reduction in other languages; word-initial unstressed vowels in Russian are often reduced to a lesser degree than non-initial ones (Jaye Padgett, p.c.).

4.6 An analysis at the phonetics-phonology interface

So far, I have assumed that articulatory target regions exist and that they are associated with different sound categories while leaving the meaning of “sound category” underspecified. In order to fully model vowel reduction in Coratino, a theory of where targets come from and how speakers choose to switch between them is needed. In particular, it must be explained why speakers tolerate the distortion of jaw/tongue body height target regions but not tongue backness target regions as *MT* decreases.

I show that a hybrid phonological-phonetic analysis based on a proposal by Eischens (2022) in combination with the Fitts’ law-based analysis sketched in this section is capable of modeling the Coratino vowel reduction patterns. In this model, the phonology provides multiple categorically differing outputs to the phonetics, and the phonetics selects from among them based on how feasibly each can be implemented in articulation based on phonetic factors such as duration.

4.6.1 Phonetic selection of phonological outputs

In his analysis of a phenomenon in San Martín Peras Mixtec in which increasing speech rate conditions a change in abstract phonological representations (specifically the deletion of a mora) Eischens (2022) proposes an explicit model of the phonetics-phonology interface, Phonetically-Informed Candidate Selection (PICS).

In the PICS model, the phonetics and phonology are separate modules. The phonology generates (not necessarily in real time) a ranked list of outputs (Coetzee 2006 as cited in Eischens 2022: p. 208), which are at this point still abstract

phonological representations. This list consists only of candidates which are relatively well-formed, i.e., only those which violate constraints whose rankings or weightings are relatively low.

Taking the phonology's list of outputs as its input, the phonetics then generates a mapping from each phonological output to a phonetic implementation with a concrete realization in time and space and evaluates the feasibility of each given current speaking conditions. If the realization of the highest ranked output is feasible, it is produced; if not, the realization of the next-highest ranked output is produced; and so on. In other words, if the output preferred by the phonology is ruled out in phonetic implementation, the next most preferred is attempted, and so on.

The finiteness of the ranked list of phonological outputs captures the intuition that the speaker does not go to the trouble of phonetic evaluation for candidates that score poorly in the phonology, but instead devotes their time and cognitive resources to the phonetic evaluation of a small number of relatively plausible candidates. One important consequence of this is that the phonology may output a list consisting of only one candidate, in which case this candidate must be realized no matter how egregious the phonetic constraint violations are. In §4.6.2, I argue that the realization of /a/ in unstressed, non-initial syllables provides an example of such a scenario.

A summary of a PICS/Fitts' law analysis as applied to raising and centralization in Coratino is as follows. For raising, phonological faithfulness constraints enforcing height specifications eliminate all candidates which violate them. The phonology effectively prohibits categorical neutralizations by preventing the phonetics from ever seeing candidates in which those neutralizations take place.

For centralization, the phonology is more permissive. First, the phonology provides outputs in which vowels are realized with their underlying backness specifications, and outputs in which those vowels have been centralized. Second, the phonetics chooses between those outputs based on a combination of duration and displacement criteria. Assuming a central starting position for the tongue body, the articulation of a non-central vowel requires a tongue body movement with greater D and thus greater MT than the articulation of a central vowel. Stressed and word-initial vowels have a great enough MT that it is possible to execute a tongue body movement with the required D for a non-central vowel; thus, the phonetics is permitted to select outputs with non-central vowels for implementation.

However, in non-initial unstressed syllables, the high- D tongue body movements necessary to produce a non-central vowel have an MT that exceeds the duration of the vowel. In nonprotected vowels, such movements are ruled out by Fitts' law, and the phonetics must thus select an output with a central vowel instead. In protected vowels, the longer MT of the high- D tongue body movement can be distributed across the vowel and an adjacent consonant which is also compatible with that same tongue body movement, allowing for great enough D that the phonetics can select a candidate with a non-central vowel.

In the analysis presented here, I make the following assumptions. First, that the relevant phonological features for the Coratino vowel inventory are $[\pm\text{high}]$, $[\pm\text{low}]$, and $[\pm\text{advanced tongue root}]$ ($[\pm\text{ATR}]$) for height, and $[\pm\text{front}]$, $[\pm\text{back}]$, and $[\pm\text{round}]$ for backness and roundness, and the feature specifications for each vowel are as in Table 8.

Second, I assume that each combination of features which defines a row in Table 8 corresponds to a target region for jaw/tongue body height, and that each combination of $[\pm\text{front}]$ and $[\pm\text{back}]$ that defines a column corresponds to a target region for tongue backness. (Roundness is included because it also alternates, but I do not consider target regions for the upper or lower lips in this analysis.) For ease of exposition, the canonical values for each target are stated as the mean normalized F1 or F2 value measured in stressed syllables for each height and backness category. For simplicity, from this point onward I refer simply to tongue height and backness to mean jaw and tongue body height and tongue body backness, respectively.

			$[-\text{round}]$ $[-\text{back}]$ $[\text{+front}]$	$[-\text{round}]$ $[-\text{back}]$ $[-\text{front}]$	$[\text{+round}]$ $[\text{+back}]$ $[-\text{front}]$	Tongue height target
$[\text{+ATR}]$	$[\text{+high}]$	$[-\text{low}]$	i	ɨ	u	.19
$[\text{+ATR}]$	$[-\text{high}]$	$[-\text{low}]$	e	ə	o	.25
$[-\text{ATR}]$	$[-\text{high}]$	$[-\text{low}]$	ɛ	ɜ	ɔ	.53
$[-\text{ATR}]$	$[-\text{high}]$	$[\text{+low}]$		a		.77
Tongue backness target			.73	.38	.18	

Table 8: Assumed phonological feature specifications and tongue body height and backness targets for each vowel in Coratino.

Two notes. First, $[\text{ɜ}]$ is included in Table 8 because i) it is definable via the features given here and ii) it should theoretically be achievable in articulation via a combination of the jaw/tongue body height target region of $[\text{ɛ}, \text{ɔ}]$ and the backness target of $[\text{i}, \text{ə}, \text{a}]$. However, if it did occur in the data set it was not

transcribed separately from [ə]. It is likely that at the low *MT*s necessary for centralization to occur, raising had already occurred, preventing underlying /ɛ, ɔ/ from being realized with the same height as in a stressed syllable. Consequently, while some phonological outputs are shown containing [ɜ], it should be understood that this vowel is gradiently raised to the point that it is not distinguishable from [ə].

Second, some of the underlying forms are presented with /ə/ in unstressed syllables. This is not to suggest that /ə/ is present underlyingly in Coratino, but only that the underlying quality of the vowel is unknown, so it is represented as /ə/ in lieu of the actual underlying quality. Presumably, it is underlyingly a mid vowel, or in some cases raised /a/ which has been transcribed as [ə] rather than [ɐ].

4.6.2 Harmonic Grammar and Fitts’ law analysis

Following Eischens (2022), I model both phonetic and phonological evaluation using Harmonic Grammar (“HG”; Pater 2009), in which candidate outputs are evaluated against weighted constraints. Each constraint violation is multiplied by the weight *wt* of the constraint, and the sum of all violations is assigned to the candidate as its harmony score *H*. By convention, violations are represented with a negative number to indicate dispreference. The use of HG for both the phonological and phonetic evaluations reduces the number of formalisms necessary for the model and highlights the parallels between the two evaluations.

In the phonological evaluation, the candidate with the score closest to 0 (the “most harmonic”) is **the preferred output**. The input to the phonetics is the set of phonological candidates ranked from most to least harmonic.⁵ The constraints enforce formal properties such as correspondences between input and output strings (faithfulness) or the avoidance of certain combinations of abstract features (markedness), and violations are categorical, i.e., each violation is worth -1 .

In the phonetic evaluation, the constraints enforce mappings from abstract phonological properties such as stress or vowel height to the continuous properties of target width, movement time, tongue height, and tongue backness. Because phonetic implementation is continuous, violations may be non-integer values, unlike the categorical violations counted by the phonology. A divergence

⁵While the harmony score provides a convenient way of generating a ranked list of outputs which serves as the input to the phonetics, in principle, a ranked list could also be generated using traditional Optimality Theory (Prince & Smolensky 1993/2004).

from the value demanded by a constraint results in a penalty proportional to divergence and the constraint weight, which is expressed by equation (2) where p = the penalty, wt = the weight of the constraint, and v = a value on some phonetic dimension. Values range between 0 and 1.

$$(2) \ p = wt(-1(|v_{\text{constraint}} - v_{\text{actual}}|))$$

For example, if a constraint with a weight of 3 demands that a vowel have a duration of at least .50 units, and the vowel has a duration of .35 units, then the penalty is $3(-1(|.50 - .35|)) = 3(-.15) = -.45$. In both the phonological and phonetic evaluations, the cutoff at which a candidate is eliminated is a score of -1 .

Each derivation shown consists of two tableaux, one representing the phonology and one the phonetic selection process. In the phonological tableau, the vowel to be evaluated in the phonetic tableau is underlined. The phonetic tableau shows the MT , D , and W for each possible realization of the underlined sequence; this is the candidate. Each set of values satisfies the Fitts' law equation in (1) when $a = 0, b = 0.1$, but is not intended to imply exact and concrete phonetic values; they are estimates only. For readability, a transcription of each candidate is also shown, but this should be understood only as a visual aide and not as part of the candidate being evaluated.

D for each surface vowel quality was estimated using the target values given in Table 8. A Euclidean distance in normalized F1-F2 space was calculated for each pair of a vowel with a palatal, labial or velar, and coronal segmental context. The coordinates of these contexts were treated as a tongue backness target identical to a front vowel, central vowel, and back vowel, respectively, and a tongue height target of 0. Deviations from these estimated D values are shown for expository purposes.

W was not calculated, but instead estimated based on the relative widths of normalized F1 and F2 distributions. Future investigations should attempt to relate W to estimates of variance in articulation in a principled fashion. However, determining the exact relationship is beyond the scope of this paper.

I begin by discussing gradient raising and the absence of switching between targets, i.e., the absence of categorical alternations. The relevant phonological constraints for height are IDENT(high) and IDENT(low). A general schema for IDENT constraints is given in (3).

$$(3) \text{ IDENT}(F)$$

Let i be a segment in the input and o be its output correspondent. Every pair of i and o have the same specification for the feature F .

I model the absence of categorical height alternations as a combination of two factors. First, in the phonology, IDENT(high) and IDENT(low) are highly weighted, meaning that candidates with unfaithful height specifications fall below the cutoff and are eliminated. Second, in the phonetics, a constraint MAP(low) prevents the gradient raising of vowels specified [+low] beyond a certain threshold. Because normalized F1 is treated as a proxy for tongue height, a tongue height of 0 represents full closure, while a tongue height of 1 represents full opening.

(4) MAP(low)

A [+low] vowel is realized with a tongue height of .60 or greater.

The other relevant phonetic constraint is DUR($\check{V}$), which requires unstressed vowels to have a duration (but not necessarily a movement time) of .10 units or less. A more complex model of the phonetics would model the duration of individual segments as a function of stress, phrase-level prosody, and sentence-level prosody, as well as the demands of individual articulatory gestures that must satisfy Fitts' law. For now, this constraint describes the tendency of unstressed vowels to be short.

(5) DUR($\check{V}$)

Vowels in unstressed syllables have a duration of .10 or less.

Although unstressed [a] is long enough to incur penalties from this constraint, MAP(low) is weighted more heavily, and requires that [a] be realized with a D that results in it having a duration greater than .10 units. In this case, the duration of the vowel is equal to its MT .

(6) [tap'pa] 'plug' Phonology

/tap'pa/	IDENT(low) $wt = 1$	H
1. $\text{t}\underline{\text{a}}\text{p}'\text{pa}$		0
2. $\text{t}\underline{\text{ə}}\text{p}'\text{pa}$	-1	-1

Phonetics

1.	[tap'pa]	MAP(low) <i>wt</i> = 10	DUR($\check{V}$) <i>wt</i> = 8	<i>H</i>
	<i>MT</i> = .3 1a. [e] <i>D</i> = .6 <i>W</i> = .15		-.2	-1.6
	<i>MT</i> = .1 1b. [ə] <i>D</i> = .35 <i>W</i> = .35	-.25		-2.5

Word-initial lengthening is modeled with the constraint **DUR(#V)**. This constraint is weighted more heavily than DUR($\check{V}$) with the consequence that word-initial vowels are longer than other unstressed vowels and thus undergo an intermediate degree of raising.

(7) DUR(#V)

Vowels in word-initial position have a duration of 0.25 or greater.

(8) [ɔrdə'na] ‘ranger’

Phonology

/ɔrdə'na/	IDENT(high) <i>wt</i> = 10	<i>H</i>
1. ɔrdə'na		0
2. urdə'na	-1	-10

Phonetics

1.	ɔrdə'na	DUR(#V) <i>wt</i> = 10	DUR($\check{V}$) <i>wt</i> = 8	<i>H</i>
	<i>MT</i> = .26 1a. [ɔ] <i>D</i> = .45 <i>W</i> = .15		-.16	-1.28
	<i>MT</i> = .1 1b. [o] <i>D</i> = .15 <i>W</i> = .15	-.15		-1.5

In contrast with raising, centralization involves phonetic selection of categorically different phonological candidates. The relevant phonological constraints for centralization is IDENT(front) and IDENT(back), which together describe the

phonology’s preference for underlying backness specifications to survive. However, they are not weighted so heavily that centralization eliminates a candidate from the running.

The relevant phonetic constraint, DUR(‘V), describes the tendency of stressed vowels to have long durations. Stressed syllables are long enough that the faithful candidates are not eliminated by the phonetics; being higher ranked, they are always selected. This example also illustrates how *W* at longer movement times can shrink dramatically.

(9) DUR(‘V)

Vowels in stressed syllables have a duration of 0.50 or greater.

(10) [‘limə] ‘lime’

Phonology

/limə/		IDENT(front) <i>wt</i> = .1	<i>H</i>
1.	‘limə		0
2.	‘limə	−1	−.1

Phonetics

		1. ‘limə 2. ‘limə	DUR(‘V) <i>wt</i> = 1	<i>H</i>
☞	1. [‘limə]	<i>MT</i> = .53 <i>D</i> = .4 <i>W</i> = .02		0
	2. [‘limə]	<i>MT</i> = .42 <i>D</i> = .19 <i>W</i> = .02	−.08	−.24

When the *MT* of the tongue body backness gesture is great enough, the front and back target regions corresponding to these feature specifications can be reached, and are. However, when *MT* is too short, the phonetics rules out the first candidate and instead implements the second, which contains a central vowel whose corresponding tongue backness target region can be reached in the given *MT*. This is what occurs with unstressed nonprotected vowels.

(11) [li‘matə] ‘limed’

Phonology

/limatə/	IDENT(front) <i>wt</i> = .1	<i>H</i>
1. li'matə		0
2. li'matə	−1	−.1

Phonetics

	1. li'matə 2. li'matə	DUR($\check{V}$) <i>wt</i> = 8	<i>H</i>
	<i>MT</i> = .24		
1. [i]	<i>D</i> = .4 <i>W</i> = .15	−.14	−1.12
 2. [i]	<i>MT</i> = .13 <i>D</i> = .19 <i>W</i> = .15	−.03	−.24

A highly-weighted phonological markedness constraint $*[+round]/[-back]$ eliminates central rounded vowels before they reach the phonetics. IDENT(round) is not weighted heavily enough to prevent categorical unrounding. A general schema for context-free markedness constraints is given in (12).

(12) $*[F]/[G]$

No output segment simultaneously has the feature specifications $[F]$ and the feature specification $[G]$.

For space, the phonetic selection of [abbri'fa], which is exactly parallel to the selection of [li'matə] 'limed' in (11), is not shown.

(13) [abbri'fa] 'to burn'

Phonology

/abbru'fa/	$*[+rd]/[-bk]$ <i>wt</i> = 1	IDENT(back) <i>wt</i> = .1	IDENT(rd) <i>wt</i> = .1	<i>H</i>
1. abbru'fa				0
2. abbri'fa		−1	−1	−.2
3. abbr̥u'fa	−1	−1		−1.1

Unstressed protected vowels differ in that their *MT* is the sum of their own duration plus the duration of a neighboring consonant. This allows them to satisfy DUR($\check{V}$) while having a high enough *MT* for their gestures to reach non-central target regions.

(14) Phonology

/mol'lettə/	IDENT(back) $wt = .1$	IDENT(round) $wt = .1$	H
1. mol'lettə			0
2. məl'lettə	-1	-1	-.2

Phonetics

1. <u>mol</u> 'lettə 2. mə <u>l</u> 'lettə		DUR($\check{V}$) $wt = 8$	H
1. [mo]	$MT = .17 = .1_m + .07_o$		0
	$D = .25$		
	$W = .15$		
2. [ə]	$MT = .21$ $D = .32$ $W = .15$	-.11	-.88

The intuition that vowel reduction is blocked when a place of articulation is shared between a consonant and vowel is not novel. D'Introno & Weston (2000) propose an analysis in which centralization is modeled as the deletion of underlying backness features in unstressed vowels. This deletion is blocked when the backness features can link to an adjacent segment, i.e., a consonant with the same place of articulation. Bucci (2013a,b, 2018) proposes an analysis that is similar in spirit in which phonological representations contain a timing tier and vowels must link to at least two positions on this tier in order to be realized with their underlying backness intact. Such linking is only possible when a free position on the timing tier is available for the vowel to associate with, which occurs in stressed syllables or at the word edge, or if the adjacent consonant shares its place of articulation with the vowel, in which case the vowel is permitted to link to that consonant's position.

While these analyses successfully describe the blocking effect, they do not explain why such a state of affairs should occur. The current analysis instead treats centralization as conditioned entirely by the movement time of the vowel's articulatory gesture, and attributes the blocking effect to increases in effective MT afforded by adjacent consonants sharing a target region with the vowel. By relating target overlap, displacement, and movement time, the PICS/Fitts' law model begins to move beyond description and towards an explanation.

I have argued that the coexistence of gradient raising and categorical centralization in Coratino can be explained by a model of articulation in which

articulatory gestures obey Fitts' law. In this model, changes in duration condition gradient changes in vowel quality. If the phonology offers categorically differing alternatives to the phonetics, duration also conditions the selection of those alternatives if the phonology's most preferred candidate is not feasible to implement in articulation.

One may argue that the PICS/Fitts' law analysis is more complex than necessary to describe the Coratino facts. Purely phonetic or purely phonological accounts might indeed be simpler, but they also have shortcomings of their own.

4.7 Alternative approaches

Vowel reduction is a case in which the parallelism between phonetic and phonological processes is apparent. Accounts of vowel reduction must attempt to explain why this parallelism arises, and the cases in which it breaks down. Such accounts may divide explanatory labour differently between the phonetics and the phonology, or reject the phonetics-phonology division entirely. To understand how these accounts differ from the one presented here, it is useful to ask, How does each account explain why vowel reduction in Coratino differs from more typologically common patterns of vowel reduction?

To summarize what must be explained, vowels in unstressed syllables are subject to gradient raising proportional to their duration; shorter vowels are raised more. Raising does not result in categorical neutralization of height contrasts; even in the shortest vowels, three height categories survive. Non-initial unstressed vowels are subject to categorical centralization and unrounding which neutralizes the contrast between front unround and back round vowels. This process is blocked if the vowel is adjacent to a consonant with which it shares a place of articulation (palatal consonants for front vowels, labial and velar consonants for back vowels).

Almost all patterns of vowel reduction in unstressed syllables involve the neutralization of height contrasts; for backness contrasts to be categorically neutralized while height contrasts are not is extremely rare or even unattested, and is typologically unexpected (Barnes 2006: pp. 16, 72). But this is precisely what is found in Coratino, and is the key fact that any analysis of Coratino vowel reduction must address.

Phonological accounts place the explanatory burden for cross-linguistic differences in vowel reduction primarily on the phonology, and do not consider speaker control over phonetics as an explanatory mechanism. For example,

(Crosswhite 2000, 2004) proposes that phonological constraints which lead to vowel reduction arise from universal phonetic constraints on the articulation and perception of vowels in unstressed syllables. The action of these constraints results in a phonology which is only capable of representing typologically “natural” processes (Hayes 1999).

In a (primarily) phonological account, the coexistence of gradient raising and categorical centralization is relatively straightforward to model. Gradient raising is assumed to be phonetic, and is not represented in the phonology, while categorical centralization is modeled in the phonology. In Crosswhite’s approach, centralization of unstressed vowels may be modeled as the positional neutralization of backness and roundness contrasts (Beckman 1997), while the resistance of initial vowels to centralization may be modeled as positional faithfulness (Beckman 1998). The survival of backness contrasts in protected vowels requires some additional theoretical machinery, such as the approaches by D’Introno & Weston (2000) and Bucci et al. (2019) discussed in §4.6.2, but this is not a problem. The one area in which a primarily phonological account faces difficulties is in explaining the resistance of /a/ to raising and reduction of its duration. If concrete factors such as the duration and jaw/tongue body height of a vowel are to be modeled as automatic and not under speaker control, then the unusual behavior of /a/ must somehow be construed as automatic.

It is likely that these difficulties are surmountable. However, in the face of cross-linguistic evidence for speaker control over phonetic realizations, and given that it is possible to model the interaction between phonetics and phonology at the interface between them (Eischens 2022), we must ask what advantages the primarily phonological approach offers. It is formally simpler than the interface approach, requiring only a single set of computations to be modeled for each output, but this simplicity is possible because it lacks an explicit model of the phonetics-phonology interface. Without such a model, appeals to the phonetics cannot progress beyond the noting of parallelisms between phonetic and phonological processes.

Ohala (1981, 1993), Lindblom (1990), Lindblom et al. (1995), and Barnes (2006) all propose models in which phonological patterns are shaped over time by inputs which reflect phonetic processes. In essence, all propose that (contra Hayes 1999) the phonology does not directly concern itself with phonetic naturalness, but merely that phonological generalizations reflect patterns which happen to arise from phonetic processes. While such models offer an explanation of how vowel reduction might arise over generations of language transmission

or even within an individual language user’s lifetime, they do not explain synchronic relationships between concrete, online factors such as the duration of individual vowels and categorical changes such as centralization in Coratino.

One way in which the need for a model of the interface might be obviated is to reject a distinction between phonetics and phonology and treat them as a single module. Flemming (2001) proposes that both gradient and categorical phenomena can be represented in a single quantitative model, and that there is no need to distinguish phonetic and phonological representations. In this model, categoricity derives from perceptual contrasts; thus, if two categories are neutralized in articulation, it is because they cannot be perceptually distinguished.

An analysis of vowel reduction in this model treats categorical alternations between vowel inventories as the result of speakers attempting to find vowel inventories which are maximally perceptually dispersed (Liljencrants & Lindblom 1972) within a shrinking vowel space (Flemming 2005). Categorical raising, for example, is modeled as the F1 dimension of the vowel space compressing to the point that the maximum number of vowel categories which can fit within the space reduces, resulting in neutralization. Categorical centralization would thus be modeled as the F2 dimension compressing to the point that the most dispersed set of vowels is a vertical system [i, ə, ʊ].

In principle, this model could describe vowel reduction in Coratino. It is likely that arguments discriminating between a non-modular analysis such as the one proposed by Flemming (2001, 2005) and a modular analysis with interactions at the interface such as the PICS model (Eischens 2022) will rely more heavily on broad theoretical desiderata than on the details of any individual phenomenon. For now, I leave this as an open question.

5 Conclusion

Connecting phonological generalizations with phonetic ones requires an explicit, predictive model of the phonetics-phonology interface. This in turn requires explicit models of constraints on the phonetics, e.g., Fitts’ law, and how these constraints create principled gaps in phonetic substance (Lindblom 1972, Ohala 1983, Stevens 1998, Turk & Shattuck-Hufnagel 2020, i.a.). These models allow us to better understand the distribution of properties such as categoricity and gradience, sensitivity to phonetic factors such as surface duration or displacement between individual articulatory gestures, speaker control and language-

specific phenomena, and isomorphism between phonetic and phonological processes, or lack thereof.

In this paper, I have attempted to connect two patterns of vowel reduction in Coratino, one gradient and one categorical, to the same phonetic factor, vowel duration. In doing so, I argued that articulatory gestures obey Fitts' law, a proportional relationship between a movement's duration, displacement, and accuracy. I used the PICS model (Eischens 2022) to model the phonetics evaluation of the realizations of phonological outputs which Fitts' law permits; in the process, duration, displacement, and target width influence the chance of categorical changes in vowel quality occurring. This explicit model of the phonetics-phonology interface and how a phonetic boundary condition can condition categorical phonological alternations is a step closer to understanding how phonological generalizations connect to phonetic ones.

Some questions remain unanswered. The assumption that the choice of realizations is random and normally distributed ignores contingencies between articulatory dimensions as well as potential contingencies between perceptual dimensions which may go on to influence articulation. For example, vowel height and nasality are neither articulatorily independent (Kunay et al. 2022 and references therein) nor perceptually independent (Beddor 1993, Kingston & Macmillan 1995, Macmillan et al. 1999, Kingston et al. 2008). In other words, *W* for movements of the velum is contingent on jaw and tongue body height, and this contingency must somehow be represented in the model. Both articulatory contingencies and covariation in production which may relate to integration or learned covariation in perception (Kingston & Diehl 1994) should be incorporated into a model of articulatory planning alongside Fitts' law.

It is unclear how well the PICS/Fitts' law approach can be distinguished from a non-modular account (Flemming 2001, 2005) on the basis of data from Coratino alone. It is likely that in-depth cross-linguistic investigations relying on careful analysis of production and perception data will be required to tease apart the predictions of a non-modular account and an account which posits interactions at the phonetics-phonology interface.

Additionally, a fuller model of articulation incorporating Fitts' law must still be developed. In this paper, normalized F1 and F2 were used as proxies for jaw/tongue body height and tongue backness, respectively, and normalized duration was used to estimate movement time. However, the application of Fitts' law to this numeric data is at best a first approximation. Articulatory studies determining the exact version of the Fitts' law equation that applies to

speech are needed, as are perceptual studies to determine whether perceptual factors contribute to vowel reduction as (Crosswhite 2000, 2004) supposes. It is unclear whether existing models of articulation such as Articulatory Phonology (Browman & Goldstein 1986, 1992) should be modified to incorporate Fitts' law, or if new models of articulation incorporating not only Fitts' law but other generalist models of human motor planning such as Optimal Control Theory (Meyer et al. 1988, Hoff & Arbib 1993, and Todorov & Jordan 2002 as cited in Turk & Shattuck-Hufnagel 2020: p. 190) should be developed from the ground up.

Finally, it is unclear whether patterns of vowel reduction other than raising and centralization should be analyzed as consequences of Fitts' law or some other boundary condition(s). In addition to raising and centralization, Crosswhite (2000, 2004) identifies a qualitatively different pattern of vowel reduction: the neutralization of contrasts between mid vowels and the corner vowels /i, a, u/ in favor of the corner vowels.

Belarusian (Czekman & Smułkowa 1988 as cited in Barnes 2006, Krivitskii & Podluzhnyi 1994 as cited in Crosswhite 2004) provides an example of a typologically unusual pattern of vowel *lowering* which results in an unstressed inventory [i, a, u]. Belarusian has five vowel qualities, /i, e, a, o, u/. In unstressed syllables, /e/ and /o/ categorically lower and centralize to [a]. There is no evidence of gradience, such as realizations of /e/ and /o/ that are intermediate in quality between their stressed realizations and [a]. Some examples of this change are given in Table 9.

Underlying	Stressed		Unstressed	
/i/	p'i'f <i>i</i> f	'you (sg.) write'	p'i'f <u>u</u>	'I write'
/u/	di'f <u>u</u>	'I breathe'	'stan <u>u</u>	'I become'
/e/	'f <i>e</i> pt	'whisper'	fap'tats' <i>i</i>	'to whisper'
	'r <i>e</i> ki	'rivers'	ra'k <u>a</u>	'rivers'
/o/	'vok <i>o</i> ni	'windows'	vak'no	'window'
	's'i i	'villages'	s'i a 'lo	'village'
/a/	'stan' <i>a</i>	'becomes'	'stan' a	'becomes'

Table 9: Obligatory, categorical merger of unstressed mid vowels to [a] and non-alternation of /i, a, u/ in Belarusian (Bethin 2012, Krivitskii & Podluzhnyi 1994 as cited in Crosswhite 2004).

Crosswhite (2000, 2004) argues that the boundary condition which leads to this pattern of vowel reduction is perceptual: listeners have difficulty discriminating between vowel qualities in non-prominent (i.e., unstressed) positions.

This results in the perceptual neutralization of all but the grossest contrasts, i.e., contrasts between the maximally dispersed (Liljencrants & Lindblom 1972, Lindblom 1986) corner vowels [i, a, u]. Learners, unable to tell that the neutralization is the result of a perceptual distortion, assume that speakers neutralize the mid vowels to [a] intentionally, and begin doing so in their own productions.

There is to the best of my knowledge no direct experimental evidence for such a boundary condition. Its existence could be experimentally tested relatively straightforwardly if it were redefined in purely phonetic terms, e.g., if the environment claimed to be antagonistic to the veridical perception of a vowel's quality were defined in terms of duration rather than prominence. While an experiment testing this claim should be carried out, it may also be possible to model a dispreference for mid vowels as arising from the combination of Fitts' law with another boundary condition: the physical limits of jaw opening.

Assuming that the jaw starts relatively closed, speakers may be able to take advantage of the physical limits of the jaw to articulate low vowels rapidly with less variance than Fitts' law would predict. It is difficult to open the jaw much further than is necessary to articulate a low vowel, meaning that as the width of a height target increases, some of it will fall outside of the physical range of jaw opening and be truncated. An analogy can be made with slamming a door open: it is easy to quickly kick a door all the way open and count on the wall to stop it, but it is very difficult to quickly kick a door precisely halfway open. Speakers might take advantage of the truncation effect by increasing the target width of low vowels in order to achieve greater displacement for the same movement time to articulate low vowels rapidly and with less variance than expected by "kicking open" their jaws. This strategy would be unnecessary for high vowels, which have small height displacements and can thus be articulated both rapidly and precisely, and it would be unavailable for mid vowels, whose targets are not truncated. If decreased displacement and increased variance (but not increased W) were heavily penalized in the phonetics, then the resulting vowel inventory might exclude mid vowels.

It is also possible that the Belarusian pattern arose not from the neutralization of existing contrasts in unstressed syllables, but rather from the failure of those contrasts to ever emerge in unstressed syllables, and that this distribution of contrasts was phonologized as the modern Belarusian alternation (Barnes 2006: pp. 36–7). It is not clear whether it is sensible to speak of the influence of a boundary condition in this case. Other typologically unusual patterns of vowel reduction seem to have been phonologized despite lacking a clear one-to-one re-

relationship with a particular phonetic boundary condition. For example, Seediq (Austronesian, Taiwan) reduces /e, o, u/ to [u] in posttonic syllables, e.g., /h-m-ajed/ → [humájuts] ‘cook (actor focus)’ vs. /haged-an/ → [hujédan] ‘cook (locative focus)’ (Barnes 2006: p. 69). The unusual alternation between [e] and [u] may have arisen through a series of diachronic steps which were individually sensible and possibly motivated by different boundary conditions.

Though it seems unlikely that Fitts’ law is the (sole) underlying boundary condition for *every* pattern of vowel reduction, I have argued that the raising and centralization of vowels in unstressed syllables should be understood as one of its articulatory consequences. I have attempted to provide a proof of concept for analyses which connect phonetic vowel reduction derived from Fitts’ law to phonological vowel reduction via the PICS model, and more generally, for analyses which connect phonetic boundary conditions to phonological processes via explicit models of the phonetics-phonology interface. More work remains to be done in extending analyses of this type to other boundary conditions and phonological processes.

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